AIML Phase 2 Report · Core Development: Payroll Anomalies, Reasons, API & Chatbot

Jaisurya Marisamy | AIML track | Phase 2 | All companies and datasets in the programme (Aura Workforce, Growmar, Zenith Nova) are fictional.

1. Student and Project Details

Name	Jaisurya Marisamy	Employee ID	GM00134
Track / Phase	AIML / Phase 2	Git tag	phase-2
Project	Core Development: payroll outlier detector, reason codes, anomaly API, HR chatbot (Tasks 6 to 10)	Programme / company	AI/ML Engineer - Growmar
GitHub repository	https://github.com/jaisurya04/cortexhr (folder: phase2/)	Dataset	CortexHR_Dataset.xlsx (fictional)

2. Phase 2 Tasks and Evidence

#	Task name	Status	Evidence (paths relative to the phase2/ folder of the repository)
6	Catch payroll mistakes: payroll outlier detector	Completed	src/cortexhr/phase2/payroll.py; notebooks P2_T06 and 07 to 11; reports/phase2/payroll_cx06_per_type.csv (8 error types); rules + IsolationForest vs baseline precision 0.217 / recall 0.400
7	Explain every flag: reason codes and money at risk	Completed	reports/phase2/payroll_anomaly_explanations.csv and payroll_review_sample_20.csv (20 rows, official columns, ranked by money_at_risk); notebook 12
8	Your first API: the anomaly service	Completed	05_Starter/api_skeleton/app/main.py; models/ + model_registry.json; tests_phase2/test_api.py (10 tests incl. 422 bad input); curl commands in README.md
9	Teach the machine to understand HR questions	Completed	src/cortexhr/phase2/chatbot.py; chatbot_metric_summary.csv; chatbot_confusion_matrix.csv/.png; accuracy 0.8106 (107 of 132)
10	Slots, memory and permissions	Completed	chat_transcripts.md (multi-turn chats + refusal tests); chatbot_dialogue_results.csv (12 dialogues, 28 turns, 9 real service calls)

3. Proof That the Project Works

Component	What was done (actual)	Result / evidence
Payroll features and rules	Ratio features (gross vs expected gross, deduction ratio, net ratio, overtime vs cap); readable rules for duplicates, negative/zero net, gross vs expected, net arithmetic, PF missing.	7,355 payslips scored; 301 flagged: 107 by rules, 194 by model only. Rules caught all 30 duplicates and all 12 gross-below-expected rows that the model missed.
Isolation Forest	IsolationForest on engineered ratios only; identifiers excluded; scaling stats saved from training so one row scores the same as in a batch; no labels used for fitting.	Measured afterwards on CX-06: combined precision 0.316 / recall 0.950; rules alone 0.860 / 0.920; model alone 0.136 / 0.300 (baseline 0.217 / 0.400). Hardest type: OVERTIME_SPIKE (recall 0.73).
Reason codes and money at risk	One reason code per kind of flag and one plain sentence with the row's own numbers; money_at_risk = gap between expected and paid; output in exactly the sample sheet columns; ranked by money at risk.	20-row clerk sample in the official 12 columns (test passes). Friend test: [friend name and result to be filled in].
Anomaly API	FastAPI + Pydantic: POST /v1/anomaly/payroll and /attendance (+ batch, 1 to 500 rows), /v1/chat, /v1/health; models loaded once in lifespan; unknown fields and bad ranges return 422.	10 API tests written (6 required). Full suite on my laptop with FastAPI installed: pytest -q tests_phase2 = **69 passed** (59 service-layer + 10 API tests); together with the 32 Phase 1 tests: 101 passed.
Intent chatbot	TF-IDF (word 1-2 grams + character 2-5 grams) + Logistic Regression, 858 messages, 33 intents; FAQ answers from hr_faq; fallback below confidence 0.30 with HR contact.	Held-out accuracy 0.8106 (107 of 132, 25 errors); fallback threshold chosen on training data only (5-fold CV: 92% answered, 100% accurate when answered).
Slots, memory, permissions	Month and employee slots in a session store; follow-ups ("and last month?"); calls mock HR service with X-Demo-User; role comes from the demo directory; salary_of_other always refused.	All 12 official dialogues: 28/28 turns correct intent, 9 turns made a real service call; refusal made no service call, even for HR.

Evaluating payroll detection with and without labels. Without labels I used the flag count (301 of 7,355, 4.1%), the rules-versus-model overlap, the score distribution and a 20-row clerk review. The hidden CX-06 labels were opened only after the predictions were frozen, to measure precision and recall for each of the 8 error types. Precision is low (0.316) because 206 flagged payslips are not injected errors; many are high earners the model finds unusual. I report that as a limit, not a success.

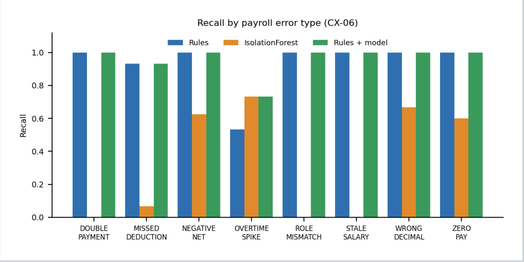


Fig. 1 Recall by error type: rules vs model vs combined (Task 6)

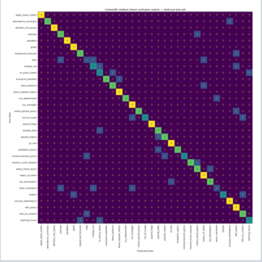


Fig. 2 Chatbot intent confusion matrix, 132 held-out messages (Task 9)

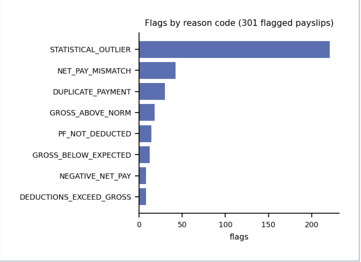
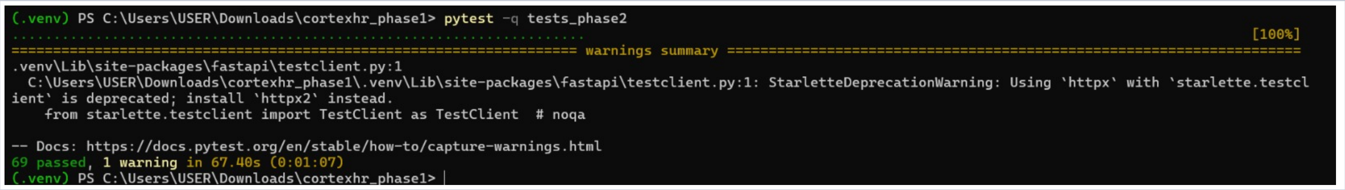


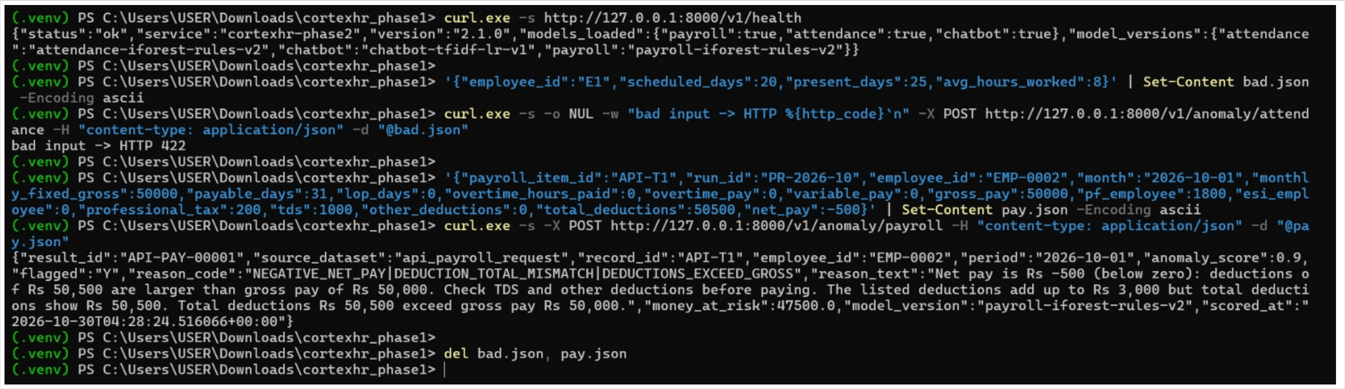
Fig. 3 Flags by reason code (Task 7)



```
(.venv) PS C:\Users\USER\Downloads\cortexhr_phase1> pytest -q tests_phase2
.....................................................................                                                         [100%]
==================================================================== warnings summary =====================================================================
.venv\Lib\site-packages\fastapi\testclient.py:1
  C:\Users\USER\Downloads\cortexhr_phase1\.venv\Lib\site-packages\fastapi\testclient.py:1: StarletteDeprecationWarning: Using `httpx` with `starlette.testcl
ient` is deprecated; install `httpx2` instead.
    from starlette.testclient import TestClient as TestClient  # noqa

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
69 passed, 1 warning in 67.40s (0:01:07)
(.venv) PS C:\Users\USER\Downloads\cortexhr_phase1> |
```

Fig. 4 Laptop screenshot: `pytest -q tests_phase2` = **69 passed** (59 service-layer + 10 API tests), 1 deprecation warning, 67.40 s.



```
(.venv) PS C:\Users\USER\Downloads\cortexhr_phase1> curl.exe -s http://127.0.0.1:8000/v1/health
{"status":"ok","service":"cortexhr-phase2","version":"2.1.0","models_loaded":{"payroll":true,"attendance":true,"chatbot":true},"model_versions":{"attendance
":"attendance-iforest-rules-v2","chatbot":"chatbot-tfidf-lr-v1","payroll":"payroll-iforest-rules-v2"}}
(.venv) PS C:\Users\USER\Downloads\cortexhr_phase1>
(.venv) PS C:\Users\USER\Downloads\cortexhr_phase1> '{"employee_id":"E1","scheduled_days":20,"present_days":25,"avg_hours_worked":8}' | Set-Content bad.json
 -Encoding ascii
(.venv) PS C:\Users\USER\Downloads\cortexhr_phase1> curl.exe -s -o NUL -w "bad input -> HTTP %{http_code}`n" -X POST http://127.0.0.1:8000/v1/anomaly/attend
ance -H "content-type: application/json" -d "@bad.json"
bad input -> HTTP 422
(.venv) PS C:\Users\USER\Downloads\cortexhr_phase1>
(.venv) PS C:\Users\USER\Downloads\cortexhr_phase1> '{"payroll_item_id":"API-T1","run_id":"PR-2026-10","employee_id":"EMP-0002","month":"2026-10-01","monthl
y_fixed_gross":50000,"payable_days":31,"lop_days":0,"overtime_hours_paid":0,"overtime_pay":0,"variable_pay":0,"gross_pay":50000,"pf_employee":1800,"esi_empl
oyee":0,"professional_tax":200,"tds":1000,"other_deductions":0,"total_deductions":50500,"net_pay":-500}' | Set-Content pay.json -Encoding ascii
(.venv) PS C:\Users\USER\Downloads\cortexhr_phase1> curl.exe -s -X POST http://127.0.0.1:8000/v1/anomaly/payroll -H "content-type: application/json" -d "@pa
y.json"
{"result_id":"API-PAY-00001","source_dataset":"api_payroll_request","record_id":"API-T1","employee_id":"EMP-0002","period":"2026-10-01","anomaly_score":0.9,
"flagged":"Y","reason_code":"NEGATIVE_NET_PAY|DEDUCTION_TOTAL_MISMATCH|DEDUCTIONS_EXCEED_GROSS","reason_text":"Net pay is Rs -500 (below zero): deductions o
f Rs 50,500 are larger than gross pay of Rs 50,000. Check TDS and other deductions before paying. The listed deductions add up to Rs 3,000 but total deducti
ons show Rs 50,500. Total deductions Rs 50,500 exceed gross pay Rs 50,000.","money_at_risk":47500.0,"model_version":"payroll-iforest-rules-v2","scored_at":"
2026-10-30T04:28:24.516066+00:00"}
(.venv) PS C:\Users\USER\Downloads\cortexhr_phase1>
(.venv) PS C:\Users\USER\Downloads\cortexhr_phase1> del bad.json, pay.json
(.venv) PS C:\Users\USER\Downloads\cortexhr_phase1> |
```

Fig. 5 Laptop screenshot, Task 8 curl evidence: GET /v1/health (all models loaded), bad attendance input returns HTTP 422, POST /v1/anomaly/payroll returns flagged Y with reason codes, plain sentence, money_at_risk and model_version.

```
You:  what is Priya Nair salary
Bot:  I can't disclose another employee's salary or payslip. You can ask about your own payroll or general salary-band policy.
      intent: salary_of_other | confidence: 1.00 | service_called: False
```

Fig. 6 Chat refusal test over HTTP (reports/phase2/chat_transcripts.md): another employee's salary is refused and no HR service call is made.

Mini challenge: how a flagged payslip travels from raw data to an API answer

1. payroll_monthly_features is loaded and ratio features are built (gross vs expected gross, deduction ratio, net ratio, overtime over cap).
2. Rules run first (duplicate, negative/zero pay, gross vs expected, net arithmetic, PF missing); IsolationForest then scores the combined ratios.
3. Each flag gets reason codes, a sentence with the row's own numbers and money_at_risk; rows are ranked by money at risk in the official columns.
4. The saved detector (models/payroll_detector_v2.joblib, loaded once) is called by POST /v1/anomaly/payroll; one row scores the same as inside a batch.
5. Real row: PI-003304 (EMP-0304, 2025-12) became result P2-00001, score 0.90, GROSS_ABOVE_NORM | NET_PAY_MISMATCH | STATISTICAL_OUTLIER, money at risk Rs 2,541,276: gross Rs 2,823,640 is 10.0x the Rs 282,364 that pay components explain (an extra digit); net pay also differs from gross minus deductions by Rs 430,107.

Think-for-yourself answers

- **Hardest error type:** OVERTIME_SPIKE (4 of 15 missed): spikes below the 50-hour cap look normal. The model alone missed all DOUBLE_PAYMENT, ROLE_MISMATCH and STALE_SALARY rows (a duplicate only shows when two rows are compared), yet a person spots a duplicate at once. Rules fix this.
- **Same reason, which first:** larger money_at_risk first, then higher anomaly_score, then earlier payroll_item_id so the order is stable.
- **Most confused intents:** help vs chitchat; the social group (greet, thanks, help, chitchat) causes 8 of 25 errors. Fix: add varied paraphrases collected outside the test set.
- **Same name:** employees only get their own data, so a request naming anyone else gets one neutral answer; HR/ADMIN get a clarifying question with employee ids and wait for one.

4. What I Learned

Rules first, model second: a rule a clerk can read beats a clever model for hard facts such as duplicates and negative pay, while IsolationForest adds review candidates the rules do not cover. A score must be scaled with statistics saved from training, otherwise one row is scored against itself. Labels measure a detector but never train it, and one overall number hides weak error types, so I measured each of the 8 types. An alert needs a reason and a rupee amount before anyone acts on it. For the chatbot, a wrong confident answer is worse than "I am not sure", and permissions must be enforced in code, not by asking nicely. My honest held-out accuracy is 81.06%, not 90%.

5. Problems Faced and Solutions

Problem	How I investigated	Solution	Result / evidence
Single-row scoring always returned 0.0	Compared one row with the same row inside a batch	Min/max scaling saved inside the detector at training time	Test: one row equals the same row in a batch
Generic reasons; money_at_risk was only abs(net pay)	Showed flags to myself as a clerk would read them	One sentence per code with real numbers; risk = gap vs expected gross	20-row sample; test: reasons never generic
Reports disagreed (0.7576 vs 0.8106 accuracy)	Compared validation summary with metric file	Script regenerates all reports; test checks they agree	test_reports_consistent passes
"September" after "Which month?" got the fallback	Ran 12 dialogues and transcripts over real HTTP	Added an awaiting-month state in the session	Dialogues: 28/28 intents; follow-ups reuse slots
Reports still said CX-06 was blocked	Searched README, notebooks and scripts for stale text	Replaced with the measured per-type table	payroll_cx06_per_type.csv; baseline comparison updated

6. Plan for Phase 3

Phase 3 moves to a new language problem: ranking resumes against job descriptions.

- Clean resume and job text; build TF-IDF similarity as the baseline and record scores.
- Explain every score in plain words (matched skills, missing skills) before showing it to a recruiter.
- Test for bias with resume_bias_pairs; measure only on held-out data and tune thresholds on a separate split.
- Raise payroll precision (206 false positives) with a separate tuning split; keep human review before any action.

7. GitHub and Submission Checklist

☑ pytest -q tests_phase2 with FastAPI installed: 69 passed.
☐ Commit all Phase 2 files; do not commit 04_Validation_Vault.
☐ git tag phase-2 and git push --tags.
☐ Verify the GitHub repository link.
☐ Friend feedback added for the 20-row test.
☑ Screenshots inserted (pytest, API curl) in section 3.
☐ Export as a single PDF with a clickable GitHub link.
☐ Upload the PDF to the programme submission form.